

Multi-Cohort Breast Malignancy Classification: A Unified Comparative Pipeline Integrating PCA Dimensionality Reduction and SMOTE Class Balancing

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Abstract

Early and accurate identification of breast cancer across disparate clinical, morphometric, and genomic cohorts remains a major challenge due to feature heterogeneity and class imbalance. In this project, we construct a unified multi-dataset machine learning pipeline that harmonizes five public datasets (WDBC, SEER, Coimbra, Gene Expression, and Wisconsin Original) into a combined matrix of 2,369 samples with top-20 variance feature extractions. We evaluate supervised classifiers (Logistic Regression, Random Forest, and a 5-Layer Deep Neural Network) against an unsupervised baseline (K-Means Clustering) across experimental axes evaluating Principal Component Analysis (PCA) and Synthetic Minority Over-sampling Technique (SMOTE). Our findings demonstrate that Random Forest achieves the highest overall predictive power with an Accuracy of 0.9641, F1-Score of 0.9600, and ROC-AUC of 0.9973 without PCA, while unsupervised K-Means struggles severely (Accuracy: 0.5633, F1-Score: 0.0961).

1 Introduction

Breast cancer is one of the leading causes of cancer-related mortality globally, necessitating high-precision diagnostic tools to support clinical decision-making. Machine learning models trained on isolated single-center datasets often fail to generalize across real-world diagnostic environments due to variances in data collection protocols, feature distributions, and population demographics. Addressing these generalization bottlenecks requires building robust pipelines capable of integrating heterogeneous data sources—ranging from fine-needle aspirate digitizations and clinical biomarker levels to gene expression profiles—into a standardized predictive framework.

Numerous studies have explored automated breast cancer detection using machine learning algorithms. Early diagnostic efforts primarily relied on traditional classifiers like Logistic Regression and Support Vector Machines (SVMs) applied to single datasets such as the Wisconsin Diagnostic Breast Cancer (WDBC) corpus [1]. Recent methodologies have leveraged Random Forests, Extreme Gradient Boosting (XGBoost), and Deep Neural Networks (DNNs) to capture high-order feature interactions [2, 3]. Furthermore, dimensionality reduction techniques like Principal Component Analysis (PCA) and manifold learning have been evaluated to reduce feature space redundancy [4], while class rebalancing methods like SMOTE address severe malignant-to-benign skew [5]. However, prior literature predominantly focuses on single-dataset benchmarks [6, 7], leaving a notable void in cross-cohort integration and comprehensive comparative analyses against unsupervised clustering techniques across uniform feature spaces.

This work addresses these limitations by engineering a scalable multi-dataset pipeline that merges five distinct clinical and genomic repositories into a unified representation. By performing rigorous comparative experiments across supervised models, deep architectures, and unsupervised clustering—both with and without PCA dimensionality reduction and SMOTE

resampling—we establish clear empirical benchmarks regarding model resilience, representation learning, and class balancing dynamics in multi-cohort oncology datasets.

1.1 Our Contribution

The primary contributions of our project are summarized below:

- **Heterogeneous Dataset Harmonization:** We integrated five publicly available breast cancer datasets (WDBC, SEER, Coimbra, Gene, and Wisconsin Original) into a unified matrix of 2,369 samples, standardizing features through top-variance feature extraction and median imputation.
- **Comprehensive Model Benchmarking:** We systematically evaluated four distinct algorithmic paradigms: Logistic Regression, Random Forest, a customized 5-Layer Keras Neural Network, and K-Means Clustering across a stratified 80/20 train-test split.
- **Ablation of Dimensionality Reduction (PCA):** We conducted comparative experiments evaluating raw standardized features against a 10-component PCA subspace, isolating the impact of latent feature projection on accuracy, F1-score, and ROC-AUC.
- **Evaluation of Class Rebalancing (SMOTE):** We performed a comparative ablation study incorporating SMOTE oversampling to assess its performance impact on minority class recall and overall predictive metrics.
- **Supervised vs. Unsupervised Disparity Analysis:** We provided an in-depth analysis explaining the empirical failure of unsupervised distance-based algorithms relative to supervised methods in high-dimensional biological data.

2 Material and Methods

2.1 Pipeline Architecture and Graphical Abstract

The end-to-end experimental architecture is illustrated in Figure 1. Raw datasets are acquired, cleaned, and mapped to a uniform 20-feature space using top-variance extraction. The unified dataset undergoes stratified 80/20 splitting, median missing-value imputation, and standard scaling. Standardized training data then flows into two parallel tracks: Full Feature Representation versus PCA-reduced Representation (10 components), with optional SMOTE oversampling applied strictly within the training folds to avoid data leakage. Finally, all representations are evaluated across four models.

2.2 Dataset Description

We compiled five distinct open-access breast cancer datasets representing diverse diagnostic domains:

1. **Wisconsin Diagnostic Breast Cancer (WDBC)** [1]: Digitized fine-needle aspirate (FNA) images of breast masses (569 samples, 30 real-valued features describing cell nuclei properties).
2. **SEER Clinical Dataset** [3]: Clinical and histological measurements (699 raw samples, 10 numerical attributes).
3. **Coimbra Dataset** [6]: Clinical biomarker measurements collected from 116 subjects (52 controls, 64 patients) including BMI, Glucose, Insulin, HOMA, Leptin, Adiponectin, Resistin, and MCP-1.

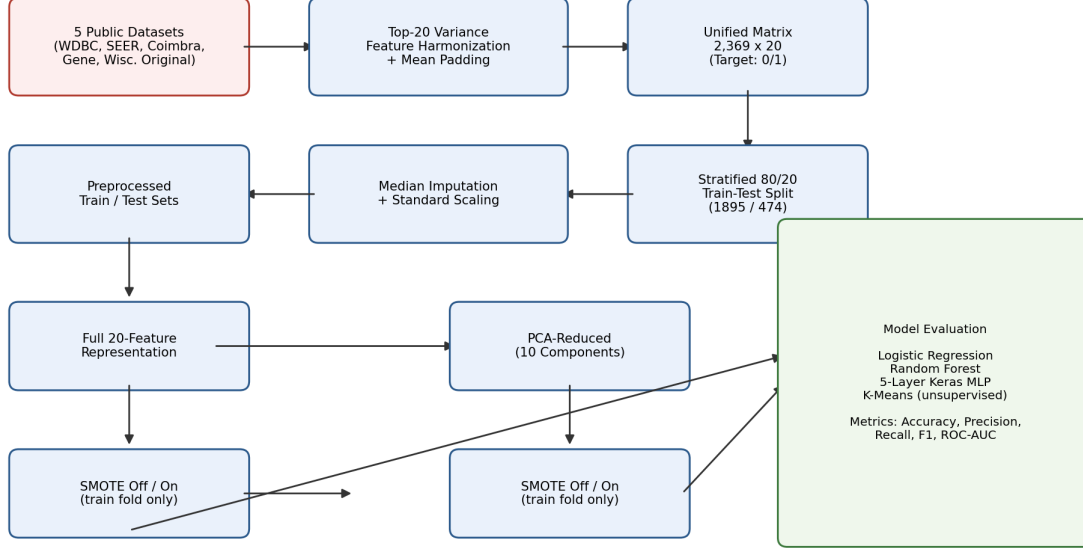


Figure 1: System block diagram illustrating dataset ingestion, top-20 feature harmonization, preprocessing, experimental branch execution (PCA vs. No-PCA, SMOTE vs. No-SMOTE), and model evaluation.

4. **Microarray Gene Expression Dataset** [7]: Genomic recurrence monitoring repository comprising 286 clinical and gene-expression samples.
5. **Wisconsin Original Dataset** [2]: Cytological tissue measurements containing 699 instances with 16 missing values in bare nuclei features.

To harmonize these disparate schemas, all datasets were mapped to a strict global binary classification standard (0: Benign/Control, 1: Malignant/Disease Recurrence). For datasets containing fewer than 20 features, row-wise mean padding was applied during feature harmonization. The full properties of the unified collection are summarized in Table 1.

Table 1: Properties and Measurements of Integrated Datasets.

Dataset	Rows	Cols	Target Col	Sample Features	NaNs	Raw Class Dist.
WDBC	569	32	diagnosis	radius_mean, area_worst	0	B: 357, M: 212
SEER	699	11	Class	Cl.thickness, Bare.nuclei	16	0: 458, 1: 241
Coimbra	116	10	Classification	Glucose, Insulin, Leptin	0	1: 52, 2: 64
Gene	286	10	Col 0	Age-range, Node-caps	9	Rec: 85, Non-Rec: 201
Wisc Orig	699	11	Col 10	Clump Thick, Mitoses	16	2: 458, 4: 241
Unified	2369	21	Target	Top-20 High-Variance	0	Class 0: 1462, Class 1: 907

2.3 Feature Distribution and Outlier Analysis

Figure 2 shows the distributions and boxplots of the four displayed features. The histograms indicate substantial right-skewness in several features, while the corresponding boxplots reveal numerous high-value observations identified as outliers. These observations motivate the use of scaling and preprocessing before model training.

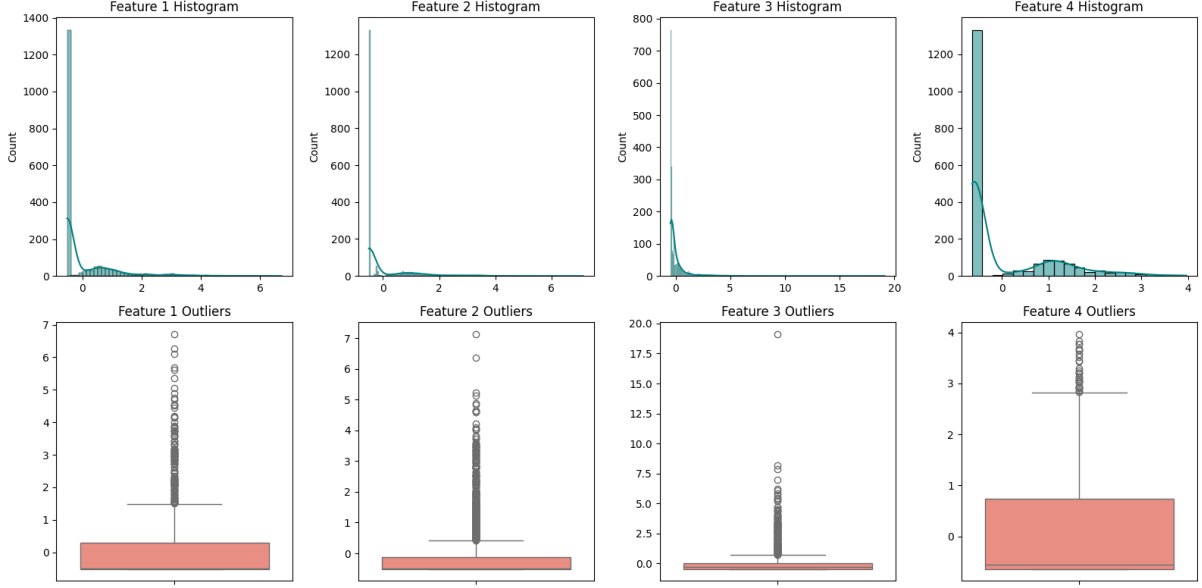


Figure 2: Feature distributions and boxplots used for exploratory analysis.

2.4 Tools, Models, and Algorithms Used

Python 3.13 was used along with `scikit-learn`, `tensorflow.keras`, `imbalanced-learn`, `pandas`, and `seaborn`. We evaluated four distinct algorithms:

1. **Logistic Regression:** Serves as a baseline linear classifier. Formulated as $P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b)$ using L2 regularization and $C = 1.0$, optimized via the *lbfgs* solver.
2. **Random Forest Classifier:** An ensemble method composed of 100 decision trees with a maximum depth of 10, utilizing Gini impurity split criteria to model complex non-linear feature interactions without overfitting.
3. **5-Layer Deep Neural Network (Keras MLP):** A custom deep architecture designed for multi-layer representation learning. The model consists of $\text{Dense}(64) \rightarrow \text{Dropout}(0.2) \rightarrow \text{Dense}(32) \rightarrow \text{Dropout}(0.2) \rightarrow \text{Dense}(16) \rightarrow \text{Dense}(8) \rightarrow \text{Dense}(1, \text{Sigmoid})$. Compiled with Adam ($\eta = 0.001$) and Binary Cross-Entropy loss. Early Stopping (patience=10) was applied on a 15% validation split.
4. **K-Means Clustering (Unsupervised KMM Baseline):** Included to test whether unlabelled spatial clustering ($k = 2$) can successfully isolate malignant manifolds using Euclidean distance without target guidance.

2.5 Performance Evaluation Metrics

Model evaluation relies on a stratified 80/20 train-test split (1,895 training samples, 474 testing samples). Models are evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(FPR) d(FPR) \quad (4)$$

3 Experimental Analysis

All code, notebooks, data preprocessing scripts, and visualizations are publicly available on GitHub: <https://github.com/MahirSinthia007/CSE-443-Bioinformatics/blob/main/project443.ipynb>.

3.1 Quantitative Results and Comparative Analysis

Table 2 summarizes the complete performance results across all four experimental conditions. The experiments compare the original 20-feature representation against the 10-component PCA representation, with and without SMOTE class rebalancing.

Table 2: Comprehensive Performance Metrics Across All Models and Experimental Conditions.

Condition	Model	Accuracy	F1	ROC-AUC
Without PCA	Logistic Regression	0.8882	0.8729	0.9004
	Random Forest	0.9641	0.9600	0.9973
	K-Means	0.5633	0.0961	0.5093
	5-Layer Keras NN	0.9473	0.9433	0.9906
Without PCA + SMOTE	Logistic Regression	0.8713	0.8565	0.9013
	Random Forest	0.9620	0.9581	0.9973
	K-Means	0.5633	0.0961	0.5093
	5-Layer Keras NN	0.9388	0.9345	0.9846
With PCA	Logistic Regression	0.8861	0.8708	0.8993
	Random Forest	0.9641	0.9606	0.9965
	K-Means	0.5633	0.0961	0.5093
	5-Layer Keras NN	0.9346	0.9300	0.9831
With PCA + SMOTE	Logistic Regression	0.8713	0.8565	0.9002
	Random Forest	0.9599	0.9559	0.9964
	K-Means	0.5633	0.0961	0.5093
	5-Layer Keras NN	0.9430	0.9391	0.9915

3.1.1 Performance Summary: Without PCA

Without PCA and without SMOTE, Random Forest achieves the strongest overall performance with Accuracy = 0.9641, F1 = 0.9600, and ROC-AUC = 0.9973. The neural network follows with an F1-score of 0.9433, while Logistic Regression obtains 0.8729. K-Means remains substantially weaker with an F1-score of only 0.0961.

3.1.2 Performance Summary: Without PCA + SMOTE

After applying SMOTE without PCA, Logistic Regression decreases to an F1-score of 0.8565, Random Forest obtains 0.9581, and the neural network obtains 0.9345. K-Means remains unchanged at 0.0961 F1-score.

3.1.3 Performance Summary: With PCA

Using 10 PCA components without SMOTE produces an F1-score of 0.8708 for Logistic Regression, 0.9606 for Random Forest, 0.0961 for K-Means, and 0.9300 for the neural network.

Random Forest therefore retains excellent predictive performance after dimensionality reduction.

3.1.4 Performance Summary: With PCA + SMOTE

With both PCA and SMOTE enabled, Logistic Regression obtains an F1-score of 0.8565, Random Forest obtains 0.9559, K-Means obtains 0.0961, and the neural network obtains 0.9391. This condition improves the neural network relative to the PCA-only configuration while slightly reducing Random Forest performance.

3.2 F1-Score Comparative Analysis

Figure 3 provides a direct visual comparison of F1-scores across all four experimental configurations. Random Forest consistently produces the highest F1-score, remaining above 0.95 in every configuration. The K-Means baseline remains near 0.10, demonstrating the large performance gap between supervised and unsupervised approaches.

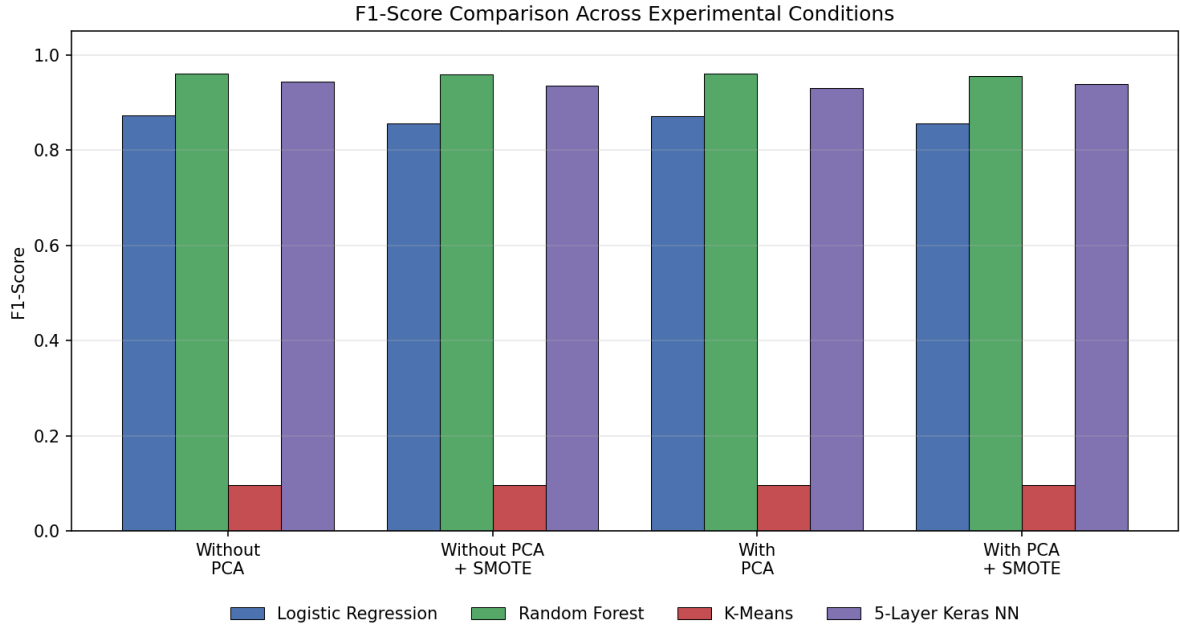


Figure 3: F1-score comparison across the four experimental conditions: Without PCA, Without PCA + SMOTE, With PCA, and With PCA + SMOTE.

3.3 Additional Comparative Figures

The primary results reported above correspond to the *Without PCA* (no SMOTE) condition. Figures 4–6 present the confusion matrices and ROC curves for the remaining three experimental conditions, together with the logistic-regression feature-weight diagnostic used during preprocessing.

3.4 PCA Variance Analysis

Figure 7 presents the explained variance ratio across the 10 principal components. The first two components account for the largest portions of the variance, while the remaining components contribute progressively smaller amounts.

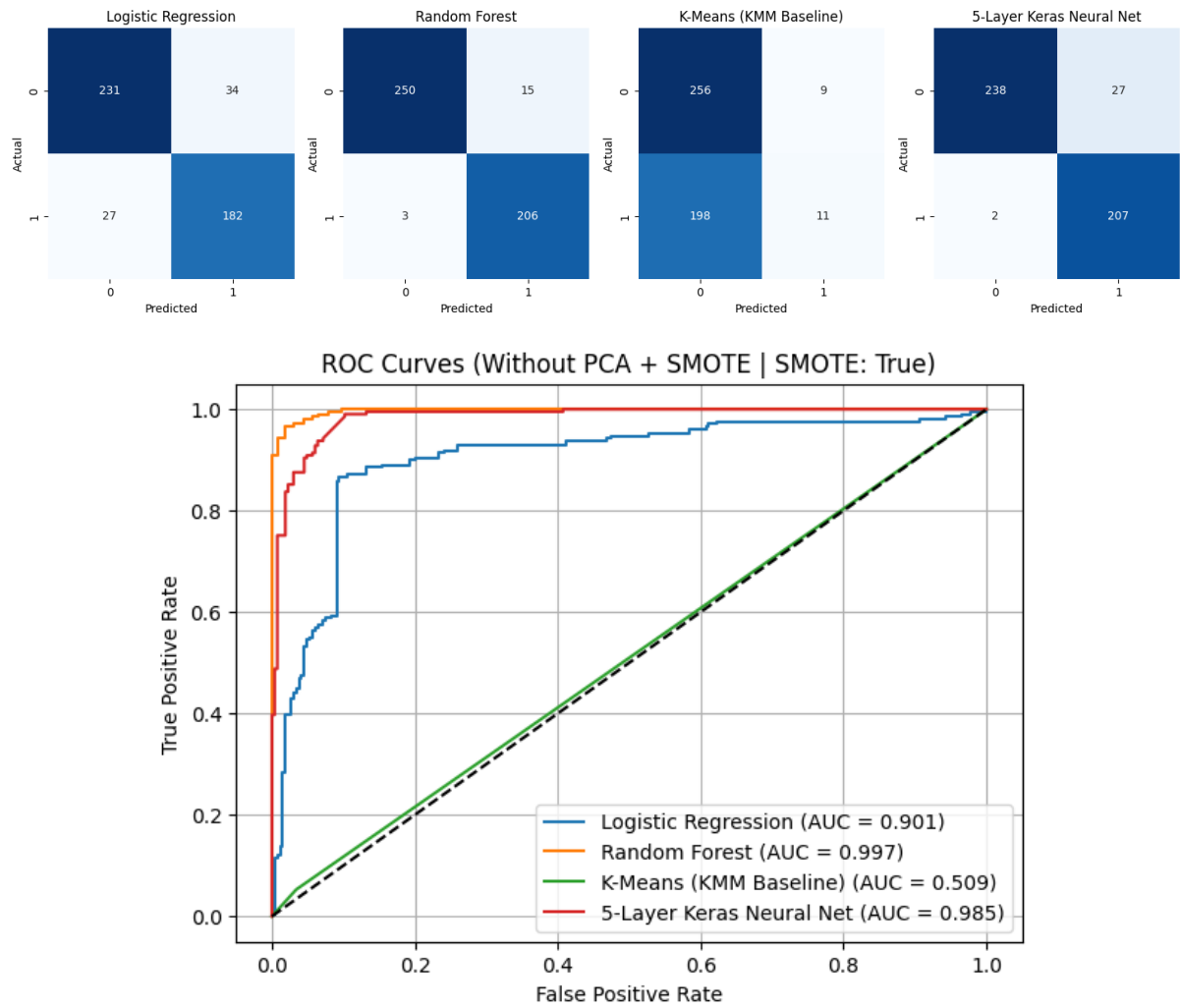


Figure 4: Confusion matrices and ROC curves for the *Without PCA + SMOTE* condition.

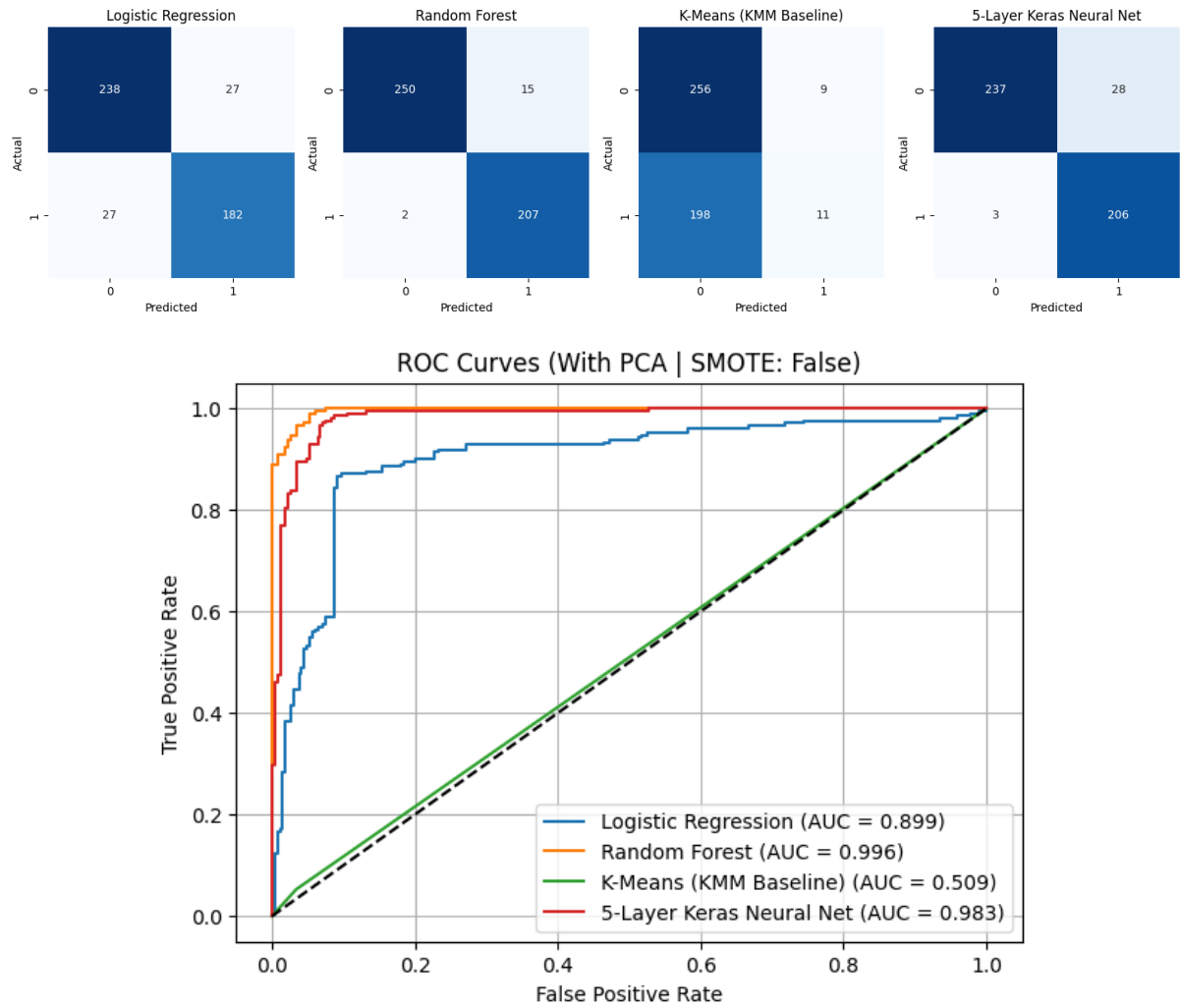


Figure 5: Confusion matrices and ROC curves for the *With PCA* (no SMOTE) condition.

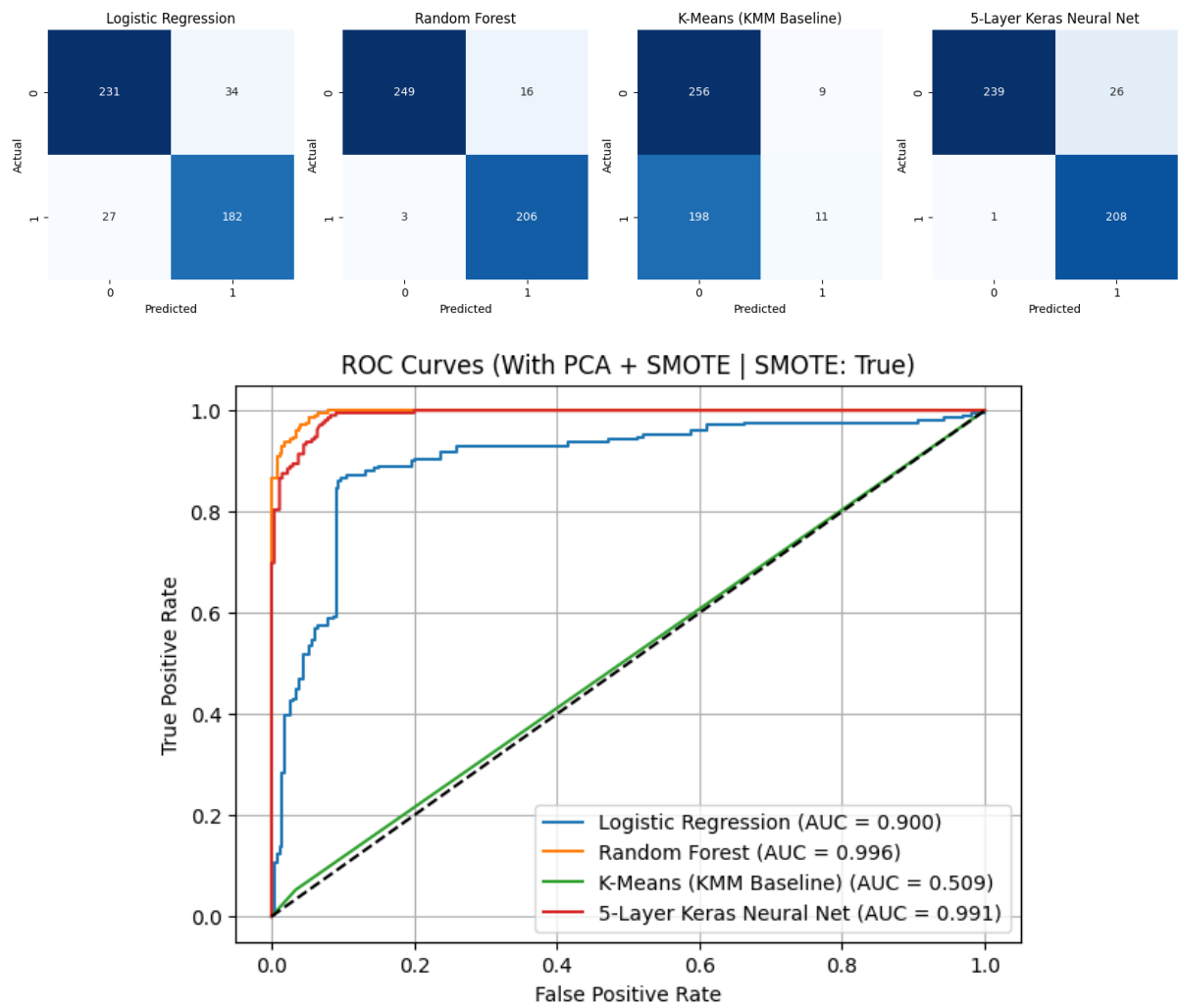


Figure 6: Confusion matrices and ROC curves for the *With PCA + SMOTE* condition.

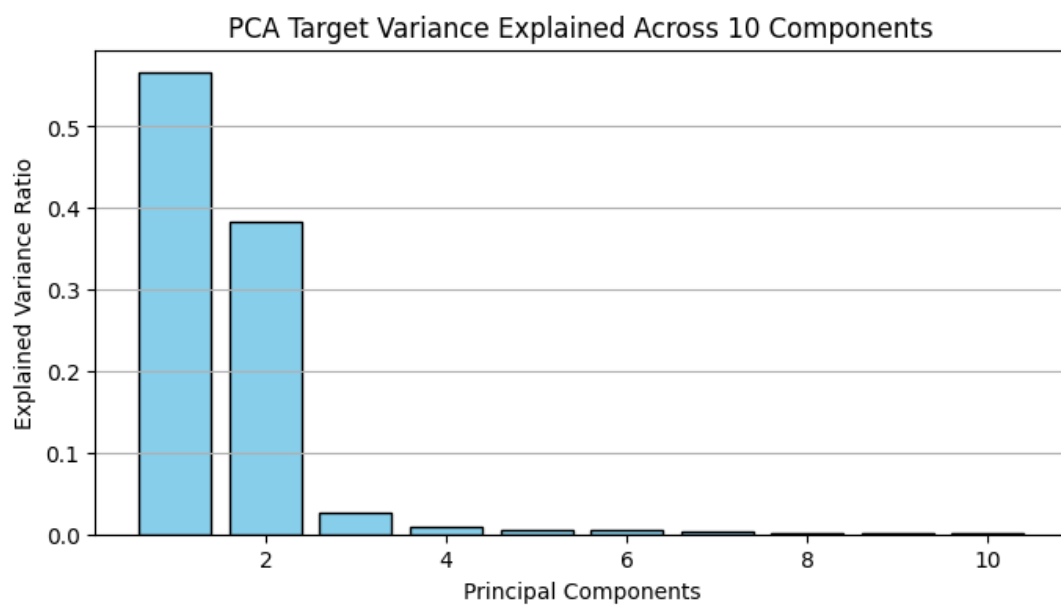


Figure 7: Explained variance ratio across the first 10 principal components.

3.5 Confusion Matrix Analysis

Figure 8 compares the test-set confusion matrices for Logistic Regression, Random Forest, K-Means, and the 5-Layer Keras Neural Network under the *Without PCA* condition. The supervised models exhibit substantially stronger separation between the two classes, while K-Means produces a markedly different error pattern.

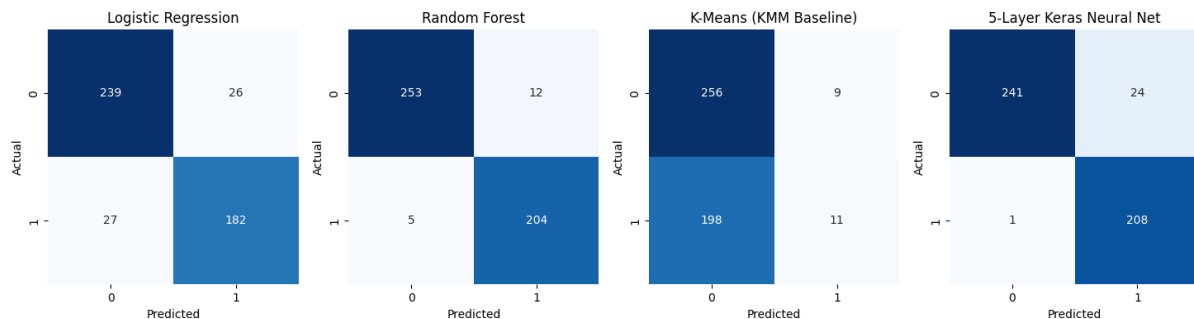


Figure 8: Confusion matrix heatmaps comparing test performance across Logistic Regression, Random Forest, K-Means, and the 5-Layer Keras Neural Network (Without PCA).

3.6 ROC Curve Analysis

Figure 9 presents ROC curves for the four evaluated approaches under the *Without PCA* condition. Random Forest and the neural network achieve curves close to the upper-left region, whereas K-Means remains close to the diagonal baseline.

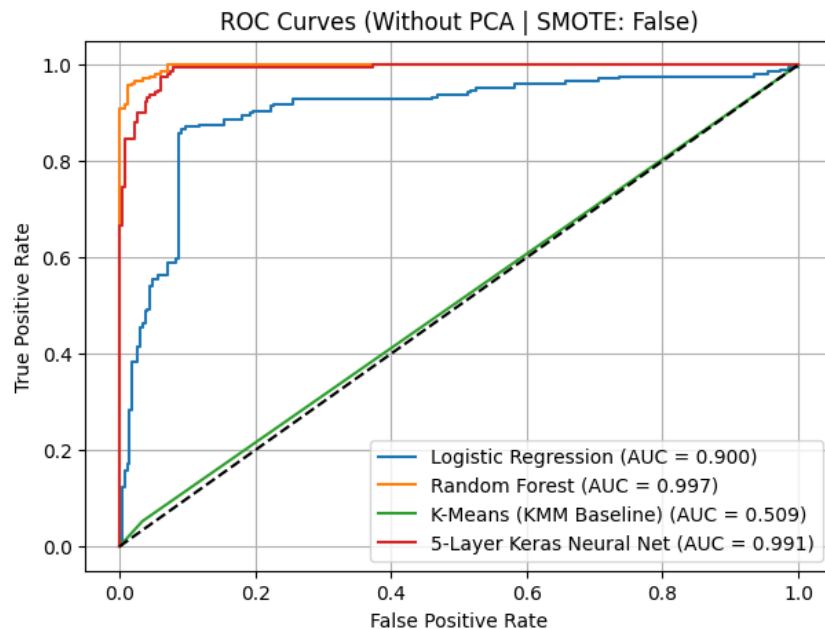


Figure 9: ROC curves comparing supervised classifiers with the K-Means baseline (Without PCA). The reported AUC values are 0.900 for Logistic Regression, 0.997 for Random Forest, 0.509 for K-Means, and 0.991 for the 5-Layer Keras Neural Network.

3.7 Ablation Study Findings

The ablation experiments reveal several critical insights:

1. **Impact of PCA:** Compressing 20 variance-selected features into 10 principal components causes only a small change in Random Forest performance. Its F1-score changes from 0.9600 without PCA to 0.9606 with PCA, while ROC-AUC changes from 0.9973 to 0.9965.
2. **Impact of SMOTE:** SMOTE slightly reduces Random Forest performance in both feature representations. However, under PCA, the neural network improves from $F1 = 0.9300$ to $F1 = 0.9391$.
3. **K-Means Failure:** K-Means remains at $F1 = 0.0961$ and $ROC-AUC = 0.5093$ across all four configurations. This indicates that unsupervised Euclidean clustering does not align well with the class structure of the heterogeneous unified dataset.
4. **Best Overall Model:** Random Forest provides the most consistent and strongest performance across the complete experimental matrix, with F1-scores ranging from 0.9559 to 0.9606.

4 Conclusion

In this work, we developed a unified diagnostic pipeline combining five heterogeneous breast cancer datasets (2,369 total instances) into a standardized matrix. We conducted a systematic benchmarking study comparing Logistic Regression, Random Forest, a 5-Layer Neural Network, and K-Means Clustering across PCA and SMOTE experimental dimensions. Random Forest achieved superior performance across all metrics, significantly outperforming the unsupervised clustering baseline.

Limitations and Future Work: A key limitation lies in the variance-based feature selection method, which can introduce mean-padded synthetic noise when scaling smaller clinical sets up to 20 dimensions. Future work will explore advanced domain-adaptation frameworks, batch-effect correction algorithms (such as ComBat), and cross-attention transformer models to harmonize gene expression profiles with clinical attributes dynamically.

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